Huawei Access Node KR-ACC-12 SOP for Minor Hardware/Environmental Fault (ALARM_4493)

Node: KR-ACC-12 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_4493

Description

The failure domain of ASIC memory parity errors, line card crashes, and power supply redundancy loss on Huawei Access Node KR-ACC-12 is primarily attributed to the complex interplay of high-speed switching fabrics and thermal management systems. The Huawei Access Node KR-ACC-12 employs a distributed architecture, where multiple line cards are connected through high-speed interconnects to form a single logical switching unit. In the event of a hardware failure or environmental anomaly, the system's distributed monitoring and control mechanisms trigger the ALARM_4493 alarm code, indicating a minor hardware or environmental fault. This fault can be caused by a variety of factors, including but not limited to, ASIC memory parity errors, line card crashes, or power supply redundancy loss. It is essential to quickly identify the root cause of the failure to prevent further system degradation and maintain optimal network performance.

Pre-requisite Checks

- Ensure the remote hands are dispatched to the KR Puram Access site to provide physical access to the Huawei Access Node KR-ACC-12.

- Verify the status of the redundant link to ensure it is operational and not contributing to the failure.

- Check the environmental conditions at the site, including temperature and humidity levels, to rule out any thermal or environmental-related issues.

Potential Root Causes

Cause	Probability
ASIC memory parity error	High
Line card crash	Medium
Power supply redundancy loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display fault-manage fault-info, Expected Output: Fault Information:, Description: This command displays the current fault information, including the fault type, location, and timestamp. It helps to identify the root cause of the failure.

Command: display system redundancy, Expected Output: Redundancy Status: Active, Description: This command displays the redundancy status of the system, including the power supply redundancy. It helps to verify if the power supply redundancy is operational.
```

```
Command: display board temperature, Expected Output: Temperature:
30°C, Description: This command displays the current temperature of
the board, which helps to identify any thermal-related issues.
```

Resolution Steps

Step 1: Step: Replace the faulty ASIC chip, Command: replace asic slot 1/0/0, Description: This command replaces the faulty ASIC chip in slot 1/0/0. It is essential to ensure the system is in a stable state before proceeding with the replacement.

Step 2: Step: Reset the line card, Command: reset line-card 1/0/1, Description: This command resets the line card in slot 1/0/1. It helps to recover the line card from a crash state.

Step 3: Step: Activate the standby power supply, Command: power supply redundancy activate, Description: This command activates the standby power supply to ensure system redundancy.

Post-Check Verification

```
Command: display fault-manage fault-info, Expected Output: No Fault
Found, Description: This command verifies that the fault has been
cleared.

Command: display system redundancy, Expected Output: Redundancy
Status: Active, Description: This command verifies that the power
supply redundancy is operational.

Command: display board temperature, Expected Output: Temperature:
30°C, Description: This command verifies that the board temperature is
within the normal range.
```

Roll-back Procedures

- Step: Restore the previous configuration, Command: undo replace asic slot 1/0/0, Description: This command restores the previous configuration and reverts any changes made during the resolution process.

- Step: Reset the system, Command: reset system, Description: This command resets the system to its previous state, ensuring all changes are reverted.

- Step: Verify system stability, Command: display system status, Description: This command verifies that the system is stable and functioning as expected.